You Can't Outsource Production Judgement

Why AI Makes Unreliable Narrators of Us All

Joe Woodhouse | DevOps Not Dead | London 2026

What We're Covering Today

Need 90 mins to do this properly.
Could rush it in 60. I have 25.

I'll go deep on what I can.
The rest is in the appendices.
You'll get the complete deck.

Act I

TL;DR

Bottom Line Up Front

LLMs are unreliable narrators

… And they make us one, too.

Individual risks vary with the amount of outsourcing

… Counter-intuitively, mid-level is the riskiest.

Risks cascade: individual to department to company.

Why You Don’t Believe Me

We feel faster with genAI LLM.
We believe you’re still doing the cognitive work.

So did they.

Why You Don’t Believe Me, cont.

METR Study (2025), gold-standard RCT

Experienced devs. Real paid work on real repos.
Subjective self-reporting: 24% faster.
Objective measurement: 19% slower.

They didn’t notice a 43% performance gap. Why?

Outsourcing Hierarchy

Level 1: AI Assist, or “Help me as I do it”

Semantic search, syntax completion

Level 2: AI Augment, or “Do some of it for me”

Prompt-by-prompt code gen, debugging, testing

Level 3: AI Outsource, or “Do it, tell me when ready”

Vibe coding, agentic, autonomous workflows

Outsourcing Hierarchy, cont.

Counter-intuitive finding, independent studies:

Level 2 is the most dangerous to individuals.

- We still engaged
- We believe we're still doing the work
- AI substitutes for our skill while we watch
- Skill decay occurs outside our awareness

Outsourcing Hierarchy, cont.

Level 3 risks are catastrophic…

but nobody's fooled about contribution

A Touchstone Test

Can we explain the answer and how it was reached
… without going back to check the LLM chat?

Yes: it’s our answer, we learned, we can recall it.

No: We used AI instead of skill.
… and we may have lost something

Act II

“Extraordinary Claims Require Extraordinary Evidence”

The Mechanisms

M1: Unreliable Narration

(M2: Skill Decay)
(M3: Security/Quality Debt)
(M4: Cognitive Overload Paradox)
(M5: Autonomous Agent Catastrophe)

Unreliable Narration - BLUF

We can’t trust LLM outputs.

“Hallucinations”; non-determinism; GIGO

We can’t trust our own perceptions of LLM use.

Contaminated planning. Metrics. Risk assessment.

This is most dangerous at Level 2: AI Augment.

We don’t notice what’s happening.

Unreliable Narration - Evidence

METR (2025): The 45% Perception Divergence

Experienced devs, real OSS repos, RCT

Claimed: 24% faster | Actual: 19% slower

Perception diverged 43% from reality

They still believed they were faster post-study!

Unreliable Narration - Evidence

MIT (2025): Cognitive Debt Study

- 54 students, 4 months, writing SAT essays
- LLM users: 83% couldn’t recall own essay
- EEG showed alpha/beta waves halved
- Neural engagement declined over time

Unreliable Narration - Evidence

Frontiers in Psychology (2025): Substitutive Offloading

- Users overestimate knowledge of AI output
- LLMs create “illusion of competence”
- Memory encoding & retrieval decline over time
- … User is unaware

Unreliable Narration - Cascade

Harms to individuals harm the Eng department

- Impact of 43% perception gap on estimation
- Project estimates driven by self-reports

Faros AI study of 10,000 devs

- Everyone claimed LLM-driven speedups
- DORA metrics flat

Unreliable Narration - Cascade

Harms to Eng department harm the whole org
- Shipping cadence
- Code quality, code effort
- AI metrics likely disconnected from P&L

We already don't usually measure what matters
- Consider AI effects on PRs, lines of code, etc.

Unreliable Narration - Mitigations

1. The Touchstone Test
   Explain the working without checking the chat?
2. Attempt the problem first before using AI
   “Desirable difficulties” create durable learning
3. Anthropic Safety Interaction Patterns
   Conceptional questioning = test score 65-86%
   Full delegation = test scores 24-39%

Unreliable Narration - Mitigations

4. No-AI Day(s)
    Compare AI-on performance to AI-off
    Similar tasks
    Track speed, errors, confidence, actual outcomes
5. Measure what matters
    Bugs in PROD, INCs (especially security), rollbacks
    NOT: commits, PRs, lines of code

Unreliable Narration - Mitigations

6. Explanation-based code reviews
   "Walk me through how this works"
   Not just "Does it past CI/CD tests?"
   Forces the dev to demonstrate understanding
7. AI-on/AI-off A/B testing at team level as well
   Measure: bugs, INCs
   Detect if "AI productivity" is eroding quality

Unreliable Narration - Mitigations

8. AI is change, so use change management
   Not: “here’s the tool, now go faster”
   But: “here’s how cognition shifts & how to detect”
9. Anthropic AI Fluency Index Training
   11 behaviours that predict effective AI use
   e.g. questioning outputs leads to better outputs

Unreliable Narration - Mitigations

10. Train your users
    ManpowerGroup: 56% received zero AI training
    Include all previous mitigations
    Show them this presentation :)

That’s All For Now

We’ve covered the most important mechanism.
There are four more in the appendices.

If You Only Remember One Thing

The Touchstone Test is at the heart of this.

Can you explain how the answer was arrived at, without going back to check the chat?

Yes: You learned, AI scaffolded your learning. Safe.

No: AI substituted for your learning. Not safe.

AI Requires Safety-Critical Thinking

Aviation doesn't ban autopilot.

It mandates minimum unaided practice.

Medicine doesn't ban diagnostic AI.

It mandates clinician benchmarking without AI.

The question isn't "Should we use AI?"

The question is "At which level, for which task, with which safeguards?"

Thank You, and Questions

Questions?

joe.woodhouse@primadonnaconsulting.com
https://www.linkedin.com/in/joewoodhouseengineer

https://github.com/Prima-Donna-Consulting/presentations/blob/current/2026.03.13.You%20Can't%20Outsource%20Judgement.Joe%20Woodhouse.DevOpsNotDead.pdf

Appendix 0x00

What Didn't Fit in 25m

TBA

[remaining mitigations added here]
[intention is that this will be complete on the day]

Appendix A

Summary of Mitigations

Mitigations: Unreliable Narration

Individual:
- Touchstone Test; conceptual questioning

Departmental:
- Measure outcomes not velocity
- Explanation-based review

Organisational:
- Close 56% training deficit
- Treat as change management

Mitigations: [TBA]

[Will add as each mechanism is added to 0x00]

Appendix B

Which AI Level, and When?

Recap: Three Levels of AI

Level 1: AI Assist (“Help me as I do it”)
Example: Syntax completion
Risk: Low (you retain cognitive load)

Level 2: AI Augment (“Do some of it for me”)
Example: “Write this function”
Risk: HIGHEST (ind), unreliable narration + skill decay

Level 3: Ai Outsource (“Do it, tell me when done”)
Example: Vibe-coding
Risk: HIGHEST (org), huge blast radius

Recap: Three Levels of AI

Level 2 is the most dangerous to individuals

We believe we’re learning; we’re not

Level 3 is the most dangerous to organisations

Trading security for velocity; cost to remediate

Use of AI Decision Tree

I have a task to complete.

Q1: Is this task security-critical?
(Authentication, authorisation, payments, privacy)

Yes: STOP. Level 0. No AI. <END>

No: <Continue to Q2>

Use of AI Decision Tree

I have a task to complete.

Q2: Do I need to learn/maintain skill in this domain?

Yes: Level 1 only (Assist). The work must be mine. <END>

No: <Continue to Q3>

Use of AI Decision Tree

I have a task to complete.

Q3: Is this high stakes if it goes wrong?
(Production, customer-facing, compliance)

Yes: Level 1-2 max (Assist/Augment).
Mandatory human review. <END>

No: <Continue to Q4>

Use of AI Decision Tree

I have a task to complete.

Q4: Can I effectively review/validate the output?
(Do I have the expertise to catch errors?)

Yes: <Continue to Q5>

No: STOP. Don’t use AI at all.
We can’t validate what we don’t understand.
<END>

Use of AI Decision Tree

I have a task to complete.

Q5: Does the task require Production access?
Or destructive actions?

Yes: Level 1-2 max (Assist/Augment), no agents.
Mandatory approval gates. <END>

No: Level 3 acceptable in sandbox/non-Prod.

Examples

Task	Recommended Level	Rationale
Learning something new	1 (Assist)	Can't outsource learning
Debugging	1-2 (Assist/Augment)	Watch for skill decay; use the Touchstone Test
Generating boilerplate	2 (Augment)	Low value, high repetition = good target for L2
Architecture design	0-1 (No AI/Assist)	High value, must retain expertise
Security code	0 (No AI)	45% vulnerabilities unacceptable
Refactoring	1 (Assist)	High skill, must retain expertise
Generating tests	2 (Augment)	Review for coverage; consider AI vs. AI
Documentation	2 (Augment)	Lower stakes but verify
Code review	0-1 (No AI/Assist)	Human in the loop critical; AI can highlight but not decide
Deploy to PROD	0-2 (No AI/Assist)	Stakes too high for L3 agents
Exploration/prototype	2-3 (Augment/Outsource)	Sandbox only; gain stakeholder agreement to not deploy

Appendix C

References, full

References

[1] METR (2025). AI Coding Assistant Productivity Study. https://metr.org/blog/2025-02-17-ai-coding-assistant-productivity-study/

[2] Purdue Skill Learning Lab (2024). AI-induced skill decay without awareness. https://cognitiveresearchjournal.springeropen.com/articles/10.1186/s41235-024-00595-1

[3] Jakesch, M., et al. (2025). Your Brain on ChatGPT: Cognitive Debt. MIT Media Lab. https://arxiv.org/abs/2501.12345

[4] Anthropic (2026). AI Assistance Impacts Coding Skills Formation. https://www.anthropic.com/research/AI-assistance-coding-skills

[5] Jose, K., et al. (2025). Outsourcing Cognition. Frontiers in Psychology. https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2025.1452345

[6] Bjork, R.A. (2011). Desirable Difficulties to Enhance Learning. https://bjorklab.psych.ucla.edu/research/

References

[7] Endsley, M.R. (1999). Level of automation effects. Ergonomics, 42(3), 462-492. https://doi.org/10.1080/001401399185595

[8] Veracode (2025). 2025 GenAI Code Security Report. https://www.veracode.com/sites/default/files/2025-genai-code-security-report.pdf

[9] Apiiro (2025). AI Code Assistants and Enterprise Security Risk. https://apiiro.com/blog/ai-code-security-risk-2025/

[10] GitClear (2025). AI Copilot Code Quality: 2025 Data Analysis. https://www.gitclear.com/coding_on_copilot_data_shows_ais_downward_pressure_on_code_quality

[11] Rezilion (2025). 30+ Flaws in AI Coding Tools. https://www.rezilion.com/blog/30-flaws-ai-coding-tools/

[12] OWASP (2025). Top 10 for LLM Applications v2.0. https://owasp.org/www-project-top-10-for-large-language-model-applications/

References

[13] NIST CVE-2025-32711: EchoLeak (M365 Copilot prompt injection). https://nvd.nist.gov/vuln/detail/CVE-2025-32711

[14] Rosen Kellerman, G. (2026). When Using AI Leads to Brain Fry. Harvard Business Review. https://hbr.org/2026/03/when-using-ai-leads-to-brain-fry

[15] Fabritz, S. (2026). AI Doesn't Reduce Work (It Intensifies It). HBR. https://hbr.org/2026/02/ai-doesnt-reduce-work-it-intensifies-it

[16] ManpowerGroup (2026). Global Talent Barometer 2026. https://www.manpowergroup.com/workforce-insights/global-talent-barometer-2026

[17] McKinsey Health Institute (2026). Employee Mental Health in Age of AI. https://www.mckinsey.com/mhi/our-insights/employee-mental-health-ai-2026

[18] Deloitte (2025). The Brain as Bottleneck: Cognitive Fatigue. https://www2.deloitte.com/us/en/insights/topics/talent/employee-cognitive-fatigue.html

References

[19] UC Berkeley (2025). AI Tools and Work Fragmentation. https://worktech.berkeley.edu/research/ai-work-fragmentation-2025

[20] Junior Developer Employment Analysis (2026). Multiple sources. Stack Overflow, Indeed, BLS data.

[21] Garman, M. (2025). AWS CEO Statement on Junior Hiring. re:Invent 2025 keynote. https://www.youtube.com/watch?v=aws-reinvent-2025-garman-keynote

[22] Anthropic (2026). Anthropic AI Fluency Index. https://www.anthropic.com/research/ai-fluency-index

[23] Karlin, B., et al. (2026). Flight Rules for Clinical AI. Nature npj Digital Medicine. https://www.nature.com/articles/s41746-026-01234-5

[24] Meta AI (2025). Agents Rule of Two: AI Agent Security Framework. https://ai.meta.com/blog/agents-rule-of-two-security-framework/

[25] NIST (2025). AI Risk Management Framework (2025 Updates). https://www.nist.gov/itl/ai-risk-management-framework

References

[26] Rubrik (2025). Agent Rewind: Immutable Audit Trail for AI Agents. https://www.rubrik.com/products/agent-rewind

[27] MIT FutureTech (2025). Agent Index: Documentation of 30 Agents. https://futuretech.mit.edu/agent-index-2025

[28] Bainbridge, L. (1983). Ironies of Automation. Automatica, 19(6), 775-779. https://doi.org/10.1016/0005-1098(83)90046-8

[29] Parasuraman, R., & Riley, V. (1997). Humans and Automation. Human Factors, 39(2), 230-253. https://doi.org/10.1518/001872097778543886

[30] Cursor IDE Incidents (March to December 2025). https://github.com/getcursor/cursor/issues/2895

[31] Replit Incident (July 2025). Database deletion, fabricated tests. Jason Lemkin Twitter: https://x.com/jasonlk/status/1946069562723897802

References

[32] Amazon Kiro Incident (December 2025, reported February 2026).
Financial Times investigation: https://www.ft.com/content/aws-kiro-incident-2026
AWS incident report: https://aws.amazon.com/message/kiro-incident-feb-2026

[33] ISO/IEC 42001:2023 (AI Management System).
https://www.iso.org/standard/81230.html

[34] EU AI Act (2024). Regulation (EU) 2024/1689.
https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689

[35] UK AI Safety Summit Commitments (2023, updated 2025).
https://www.gov.uk/government/publications/ai-safety-summit-2023

[36] Faros AI Engineering Benchmarking Report (2025).
https://www.faros.ai/blog/engineering-benchmarks-2025

Appendix D

References, by topic

References by topic

Cognitive Mechanisms (M1, M2, M4)

[1] METR (2025): 43pp perception gap study

[2] Purdue Skill Learning Lab (2024): Skill decay without awareness

[3] MIT (2025): Brain connectivity decline, cognitive debt

[4] Anthropic (2026): Skill formation inhibited by AI assistance

[5] Frontiers (2025): Substitutive offloading taxonomy

[14] HBR Brain Fry (2026): 14% brain fry, 39% more errors

[15] HBR Work Intensification (2026): Productivity treadmill

[16] ManpowerGroup (2026): Training, confidence collapse, burnout

[17] McKinsey Health (2026): Mental health decline

[23] Nature Flight Rule (2026): Aviation-medicine parallels

References by topic

Security & Code Quality (M3)

[8] Veracode (2025): 45% vulnerability rate, 100+ models

[9] Apiiro (2025): 322% priv-esc increase

[10] GitClear (2025): Refactoring collapse, duplication quadrupled

[11] Rezilion (2025): 30+ flaws in AI coding tools

[12] OWASP (2025): LLM Top 10

[13] NIST CVE EchoLeak: Prompt injection in M365 Copilot

References by topic

Agent Catastrophe (M5)

[30] Cursor incidents: Plan Mode deletion despite "read-only"

[31] Replit incident: Database deletion, fabricated tests

[32] Amazon Kiro: $47M loss, 13-hour outage, DR cascade failure

[24] Meta Agents Rule of Two

[26] Rubrik Agent Rewind

[27] MIT FutureTech Agent Index

Appendix E

About This Presentation

Constraints & Approaches

Time constraint: only 25m for main presentation.
- Deep dive on one idea. Rest into appendix.

Density choice: high information density per slide.
- Expert audience; must be its own reference.
- BLUF approach: conclusion first, then evidence.

Appendix structure
- Appendix 0x00 for unstable/overflow.
- Appendices A-G stable if overflow re-merged.
- Hex numbering visually distinct.

Constraints & Approaches

Accessibility commitment
- Text-only for screen readers
- High-contrast design, 7:1, WCAG AA compliant
- Dark mode most suitable for cinema projection

Open source commitment
- No paywalling or gatekeeping for full deck.
- Publicly available on GitHub.
- Licensed CC BY-NC-SA 4.0
- Dual-licensed for commercial use.

How To Use This Deck

For conference attendees: you saw slides 1-27
- Appendix 0x00 has what we didn’t have time for

For later readers (GitHub, LinkedIn)
- Appendix 0x00 migrating into main

For decision-making on AI use
- Start with Appendix A for Mitigation Summary
- Use Appendix B for day-to-day decision
- Measure what matters!

Appendix F

Use of AI

Use of AI in this presentation

This presentation about AI harms was created with AI assistance. Irony acknowledged.

Here's exactly how AI was used

Level 1 (Assist): Used Extensively

- Literature search
- Citation formatting
- Syntax checking and grammatical review
- Graphical design recommendations & structure

Use of AI in this presentation

Level 2 (Augment): Used Sparingly and Selectively

- Outline generation (then heavily edited by human)
- Evidence synthesis across all research
- “What have I missed here?”
- “What questions should I be asking that I didn’t?”

Level 3 (Outsourcing): Never Used

- Every slide rewritten and typed by hand
- (Except References which were copied/pasted)

Use of AI in this presentation

Why This Matters

Practice what I preach; appropriate use of AI
Retain cognitive load on synthesis and argument
Show that transparency is necessary and possible

Touchstone Test

Can I explain everything with rereading the chat?
YES. Original ideas, reviews, and edits are mine.
AI covered the field of research.

Appendix G

About the author

Joe Woodhouse

Principal Engineer | Non-Executive Director
Background in databases & consulting
Now Infrastructure/Systems/Performance Engineer

“From the server room to the board room”

joe.woodhouse@primadonnaconsulting.com
https://www.linkedin.com/in/joewoodhouseengineer/